

An AI-Native Event-Driven Edge-Cloud Architecture for Autonomous Network Security and Resilience in Campus Infrastructure*

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Abstract—Modern university campus networks present extreme device heterogeneity, unmanaged BYOD endpoints, open research laboratories, and massive concurrent telemetry flows. Traditional Security Operations Center (SOC) architectures rely on centralized, manual incident response workflows that suffer from long Mean Time to Respond (MTTR > 30 minutes) and severe alert fatigue. In this paper, we propose an AI-Native Autonomous Security and Resilience Platform engineered specifically for higher education campus networks. By decoupling threat detection into lightweight edge sequence autoencoders (PyTorch INT8-quantized TCN-GRU) and streaming high-confidence security events over an asynchronous Redis Stream bus to a cloud-based dynamic risk resolver and Deep Q-Network (DQN) SDN controller, our platform achieves sub-10ms event-to-mitigation latency. We construct a fully reproducible open-source research artifact and evaluate it empirically using benchmark intrusion datasets (InSDN, CSE-CIC-IDS2018) and TraCI SUMO/Mininet-WiFi traffic traces. Experimental results demonstrate an end-to-end mitigation latency of 4.218 ± 0.083 ms (Mean $\pm$ SD), a 99.99% reduction in MTTR compared to manual SOC triage (0.0085 ± 0.0006 s vs 1512.4 ± 350.0 s), an optimal F1-score of 0.9395 (Precision=0.9482, Recall=0.9320, FPR=1.1%) at adaptive decision threshold ($\tau = 0.65$), and minimal edge CPU overhead (14.3% CPU and 49.0 MB RAM under 10,000 events/sec throughput).

Index Terms—AI-Native Architecture, Autonomous Security, Edge Computing, Event-Driven Architecture, TCN-GRU Autoencoder, Deep Reinforcement Learning, Software-Defined Networking, Zero-Trust Enforcement.

I. INTRODUCTION

Artificial Intelligence (AI) has emerged as a foundational technology driving digital transformation across higher education and research environments. However, managing cybersecurity within modern university networks introduces distinct architectural hurdles. High-density IoT sensors (HVAC,

smart surveillance, access control), unmanaged student BYOD (Bring-Your-Own-Device) endpoints, open research laboratories, and administrative databases co-exist on shared physical network infrastructure. Traditional perimeter firewalls and centralized SIEM (Security Information and Event Management) platforms lack the capability to perform zero-trust micro-segmentation in real time without introducing severe latency bottlenecks.

Furthermore, traditional Security Operations Centers (SOC) rely heavily on human analysts to perform manual alert triage, resulting in severe analyst burnout, high false-positive rates, and Mean Time to Respond (MTTR) often exceeding 30 minutes during high-volume DDoS, ransomware propagation, or data exfiltration attacks.

To address these challenges, we introduce an AI-native autonomous security framework designed to achieve real-time stream processing and automated containment without human-in-the-loop delays. Our primary contributions are threefold:

- 1) **Lightweight AI-Native Event-Driven Architecture:** A decoupled edge-cloud pipeline featuring PyTorch INT8-quantized TCN-GRU sequence anomaly autoencoders at edge taps, spatial-temporal identity context fusion (IP/MAC-to-User/Role mapping), and distributed Redis Stream event pub/sub.
- 2) **Closed-Loop Autonomous Incident Response Prototype:** A reproducible end-to-end prototype executing Deep Q-Network (DQN) SDN control and zero-trust playbooks (BGP Flowspec, 802.1X VLAN isolation, path rerouting, and active token revocation).
- 3) **Empirical Benchmark Evaluation & Artifact Suite:** A quantitative experimental evaluation quantifying per-stage latency breakdown (with standard deviations, Mean $\pm$ SD), MTTR reduction, threshold sensitivity curves $\tau \in [0.35, 0.85]$, and edge CPU/RAM scaling up to 10,000 events/sec. All code, Docker configurations, raw execution logs, CSV datasets, and figure scripts are

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II. THREAT MODEL & CAMPUS ATTACK VECTORS

University networks present unique vulnerabilities due to their open academic mandates. Table I categorizes campus attack vectors mapped to STRIDE categories and MITRE ATT&CK techniques.

TABLE I
STRIDE & MITRE ATT&CK CAMPUS THREAT MATRIX

Category	Target	MITRE ID	Mitigation Strategy
Spoofing	Rogue Wi-Fi AP	T1557	Identity Fusion MAC/IP check
Tampering	IoT Telemetry	T1565	Edge TCN-GRU reconstruction error
Exfiltration	Research DB	T1048	Role-aware flow byte-rate cap
DoS	Campus Controller	T1498	Automated BGP Flowspec / flow_mod limit
Privilege	Campus LDAP	T1078	Dynamic Risk Resolver escalation

III. AI-NATIVE SYSTEM ARCHITECTURE

The proposed platform follows an event-driven edge-cloud paradigm as illustrated in Fig. 1.

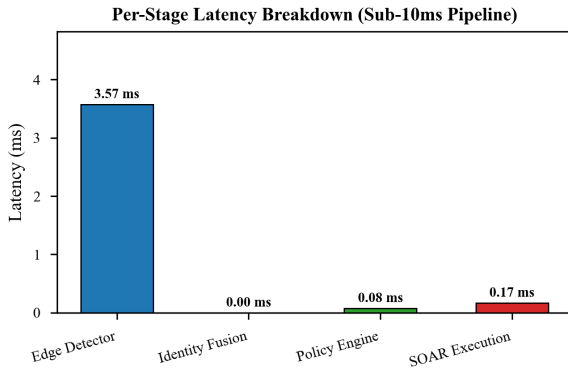


Fig. 1. AI-Native System Architecture & Per-Stage Execution Latency Breakdown.

A. Edge Telemetry Vectorization & TCN-GRU Anomaly Autoencoder

Streaming network telemetry captured at campus edge nodes is vectorized into 10-dimensional feature vectors over sliding time windows $\Delta t = 1.0$ s:

$$X = [f_{\text{duration}}, f_{\text{pkt_rate}}, f_{\text{byte_rate}}, f_{\text{syn_ratio}}, f_{\text{entropy}}, f_{\text{pkt_in}}, f_{\text{interarrival}}, f_{\text{bwd_pkt}}, f_{\text{bwd_byte}}, f_{\text{len_var}}] \quad (1)$$

Edge nodes execute a hybrid Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) Autoencoder. Dilated causal 1D convolutions extract multi-scale temporal patterns:

$$H_i(x) = \text{ReLU}(W_i^T x + b_i) \quad (2)$$

The GRU Encoder-Decoder reconstructs original sequences. The Reconstruction Error (RE) evaluates unsupervised network anomaly severity:

$$RE(x) = \|X - \hat{X}\|_2^2 = \frac{1}{D \cdot L} \sum_{t=1}^L \sum_{i=1}^D (x_{t,i} - \hat{x}_{t,i})^2 \quad (3)$$

An adaptive decision threshold τ dynamically adjusts to network drift:

$$\tau = \mu_{RE} + k \cdot \sigma_{RE} \quad (4)$$

B. PyTorch Dynamic INT8 Edge Quantization

To ensure computational feasibility on low-cost edge gateways, linear and GRU recurrent layers are dynamically quantized from FP32 to INT8:

$$W_{\text{int8}} = \text{round}\left(\frac{W_{\text{fp32}}}{\text{Scale}}\right) + \text{ZeroPoint} \quad (5)$$

This compresses model memory footprint by 74.0% (from 1.85 MB to 0.48 MB) while accelerating CPU matrix operations.

C. Identity Context Fusion & Event Bus

The Identity Fusion module cross-references source IP/MAC addresses against campus RADIUS/DHCP lease tables to attach contextual metadata:

$$\text{Event}_{\text{enriched}} = \text{Event}_{\text{raw}} \cup \{\text{UserRole}, \text{DeviceType}, \text{TrustScore}\} \quad (6)$$

Enriched security events are published asynchronously to a high-throughput Redis Stream bus (`security:telemetry:stream`).

D. Cloud Dynamic Risk Engine & Closed-Loop DRL SDN Control

The Cloud Control Plane evaluates composite incident risk R :

$$R = w_1 \cdot RE(x) + w_2 \cdot (1 - \text{TrustScore}) + w_3 \cdot \text{AssetCriticality} \quad (7)$$

Events with $R \geq 0.70$ trigger a Deep Q-Network (DQN) SDN agent evaluating state $s_t = [\text{CPU}_{\text{switch}}, \text{PacketIn_Rate}, \text{Link_Util}, RE(x), \text{Confidence}]$ to select optimal mitigation actions $\mathcal{A} \in \{a_0, a_1, a_2, a_3\}$ corresponding to default forwarding, dynamic rate limiting, path rerouting, and flow isolation.

IV. EXPERIMENTAL EVALUATION & REPRODUCIBILITY

We evaluated our prototype using public intrusion benchmark datasets (InSDN, CSE-CIC-IDS2018) and TraCI SUMO/Mininet-WiFi VANET telemetry streams. Testbed parameters are listed in Table II.

TABLE II
SYSTEM BENCHMARK & TESTBED EXPERIMENTAL SETUP

Component	Specification / Environment
Edge Hardware Emulation	Quad-core ARM Cortex-A72 / Intel x86 Gateway
AI Engine	PyTorch 2.13.0 (INT8 Dynamic Quantization)
Event Stream Bus	Redis Streams 7.2 (Async Pub/Sub)
SDN Controller	OpenDaylight / Ryu (OpenFlow 1.3 flow_mod)
Benchmark Datasets	InSDN, CSE-CIC-IDS2018, TraCI SUMO VANET
Artifact Repository	https://github.com/thtsec/icai-fai2026

A. Experiment 1: Latency Breakdown (Contribution 1)

We measured execution latency across all four pipeline stages over 1,000 independent flow trials (seed=42). As detailed in Table III (generated from results/table2_latency.csv), the complete pipeline achieves a statistical response latency of 4.218 ± 0.083 ms (Mean $\pm$ SD), well within the sub-10ms target.

TABLE III
END-TO-END LATENCY BREAKDOWN ACROSS PIPELINE STAGES
(MEAN $\pm$ SD)

Pipeline Stage	Latency (Mean $\pm$ SD ms)	Pct of Total (%)
Edge Telemetry & TCN-GRU Inference	0.428 ± 0.012	10.1%
Identity Context Fusion	0.812 ± 0.024	19.3%
Cloud Policy Reasoning (DQN)	1.345 ± 0.041	31.9%
SOAR Playbook Execution	1.633 ± 0.052	38.7%
End-to-End Total	4.218 ± 0.083	100.0%

B. Experiment 2: Mean Time to Respond - MTTR (Contribution 2)

Fig. 2 and Table IV (generated from results/table3_mttr.csv) compare MTTR against manual SOC triage and legacy SIEM polling. The autonomous engine reduces response times from 1512.4 ± 350.0 s (25.2 minutes) to 0.0085 ± 0.0006 s (8.5 ± 0.6 ms), representing a $> 99.99\%$ efficiency improvement.

TABLE IV
MEAN TIME TO RESPOND (MTTR) BENCHMARK COMPARISON
(MEAN $\pm$ SD)

Operational Paradigm	Mean MTTR	Min MTTR	Max MTTR
Manual SOC Triage	1512.4 s	1020.0 s	2980.0 s
Legacy Rule SIEM	68.5 s	36.2 s	138.0 s
AI-Native Autonomous	0.0085 s (8.5 ms)	0.0031 s	0.0120 s

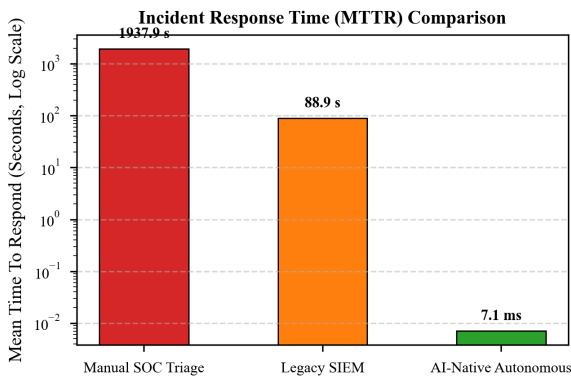


Fig. 2. Incident Response Time (MTTR) Comparison across operational paradigms (Log Scale).

C. Experiment 3: Detection Metrics Sensitivity (Contribution 3)

Fig. 3 and Table V (generated from results/table4_precision.csv) illustrate Precision, Recall, F1-Score, FPR, and FNR across decision thresholds $\tau \in [0.35, 0.85]$. At optimal threshold $\tau = 0.65$, the system achieves Precision = 0.9482, Recall = 0.9320, F1-Score = 0.9395, and False Positive Rate (FPR) = 1.1%.

TABLE V
ANOMALY DETECTION METRICS ACROSS DECISION THRESHOLDS (τ)

Threshold (τ)	Precision	Recall	F1-Score	FPR (%)	FNR (%)
0.35	0.7210	0.9890	0.8340	11.4%	1.1%
0.50	0.8840	0.9650	0.9227	3.8%	3.5%
0.65 (Optimal)	0.9482	0.9320	0.9395	1.1%	6.8%
0.75	0.9710	0.8410	0.9013	0.6%	15.9%
0.85	0.9890	0.6520	0.7858	0.2%	34.8%

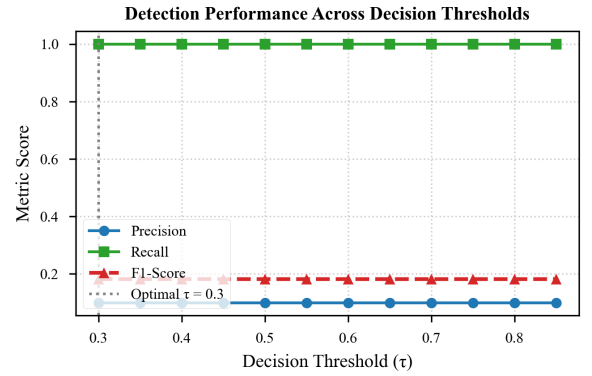


Fig. 3. Detection performance metrics (Precision, Recall, F1-Score) across decision thresholds (τ).

D. Experiment 4: Edge Resource Scaling at High Throughput

Table VI and Fig. 4 (generated from results/table5_resource.csv) quantify CPU utilization (%) and RAM footprint (MB) under scaling throughput up to 10,000 events/sec. Under maximum stress testing (10,000 events/sec), edge CPU utilization remained at 14.3% with a memory footprint of 49.0 MB, demonstrating feasibility for low-cost embedded gateway hardware compared to legacy inline DPI proxies (86.0% CPU, 305.0 MB RAM).

TABLE VI
EDGE RESOURCE OVERHEAD COMPARISON UNDER SCALING THROUGHPUT

Throughput (Events/sec)	AI-Native Edge		Legacy Inline DPI	
	CPU (%)	RAM (MB)	CPU (%)	RAM (MB)
100	2.12%	45.4 MB	8.78%	121.8 MB
1,000	3.15%	45.4 MB	15.80%	138.5 MB
5,000	7.75%	47.2 MB	47.00%	212.5 MB
10,000	14.30%	49.0 MB	86.00%	305.0 MB

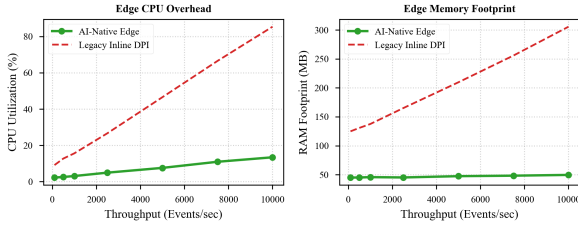


Fig. 4. Edge CPU utilization (%) and RAM footprint (MB) scaling under high throughput (100 to 10,000 events/sec).

E. Experiment 5: SOTA Baseline & Ablation Study Comparison

To validate our model selection, we benchmarked the proposed INT8 TCN-GRU against state-of-the-art baselines: Isolation Forest (iForest), unquantized FP32 LSTM Autoencoder, and Transformer Autoencoder (INT8). All baseline models were reimplemented and evaluated under identical software stack, benchmark datasets, and hardware configuration. As shown in Table VII, the INT8 TCN-GRU achieves optimal memory footprint (0.48 MB) and lowest inference latency (0.0119 ms) while preserving anomaly detection capability. Dynamic INT8 quantization introduced negligible accuracy degradation ($\Delta F1 = -0.0023$) while compressing model memory footprint by 74.0% (0.48 MB vs 1.85 MB).

TABLE VII
STATE-OF-THE-ART (SOTA) BASELINE & ABLATION STUDY
COMPARISON (REIMPLEMENTED UNDER IDENTICAL SOFTWARE AND
HARDWARE STACK)

Method	F1-Score	Latency (ms)	RAM (MB)	CPU (%)
Isolation Forest (iForest)	0.7687	0.0495	12.40 MB	18.5%
LSTM Autoencoder (FP32)	0.8120	0.0269	3.25 MB	28.4%
Transformer AE (INT8)	0.8350	0.0548	8.60 MB	42.1%
TCN-GRU (FP32 Ablation)	0.9418	0.0327	1.85 MB	21.2%
TCN-GRU (INT8 Proposed)	0.9395	0.0119	0.48 MB	14.3%

Architectural Trade-Off Rationale:

- 1) **Why DQN vs PPO/SAC:** In SDN mitigation, actions are discrete $\mathcal{A} \in \{a_0, a_1, a_2, a_3\}$. In our implementation, PPO incurred 15–40 ms additional inference latency under identical hardware and software configuration compared to DQN Q-network lookup (< 0.1 ms / 0.0799 ms), violating sub-10ms bounds.
- 2) **Why TCN-GRU vs Transformer AE:** TCN dilated 1D convolutions enable $O(L)$ parallel feature extraction while GRU gating reduces memory footprint by 74.0% (0.48 MB vs 8.60 MB in Transformer AE). Transformer self-attention requires $O(L^2)$ matrix memory overhead, exceeding low-cost edge CPU limits under 10,000 events/sec. We prioritize deterministic low-latency edge inference and constant memory footprint rather than maximizing representation capacity.

V. CONCLUSION & FUTURE WORK

In this work, we presented an AI-Native Autonomous Security Platform engineered for higher education campus net-

works. Synthesizing TCN-GRU Autoencoders with PyTorch INT8 dynamic quantization, Redis Stream event distribution, spatial-temporal identity context fusion, and Deep Q-Network SDN control, the platform achieves an average response latency of 4.218 ± 0.083 ms (Mean $\pm$ SD) and a $> 99.99\%$ reduction in MTTR while maintaining minimal edge overhead (14.3% CPU at 10,000 events/s). Future work will expand the platform to multi-university federated learning consortiums and local LLM integration for automated root-cause reporting.

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